

Music Store Analytics

Complete View Documentation – 18 Business Intelligence Views

11 Tables · 3,503 Tracks · 614 Invoices · 59 Customers

11

 Tables

18

 Views

3,503

 Tracks

614

 Invoices

59

 Customers

275

 Artists

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1

vw_invoice_detail

Master Fact Table

Business Question: What is the complete picture of every sale (each invoice line) with all related information?

Returns every invoice line item with full context: customer, employee, track, genre, artist, and album details. This is the foundation for all other analyses.

 **Tables Used:** invoice, customer, employee, invoice_line, track, genre, album, artist

 **Key Columns:** invoice_id, customer_name, track_name, genre_name, artist_name, album_title, line_total

```
-- Joins 8 tables to get full transaction context
FROM invoice JOIN customer ON invoice.customer_id =
customer.customer_id
LEFT JOIN employee ON customer.support_rep_id =
employee.employee_id
JOIN invoice_line ON invoice.invoice_id =
invoice_line.invoice_id
JOIN track ON invoice_line.track_id = track.track_id
JOIN genre ON track.genre_id = genre.genre_id
JOIN album ON track.album_id = album.album_id
JOIN artist ON album.artist_id = artist.artist_id
```

Why This Matters: Enables drill-down to any transaction.

Answers "What, Who, When, Where, How much?" – the foundation for all custom reporting.

2

vw_monthly_sales

Revenue Trend

Business Question: How is our revenue performing month over month?

Aggregates sales by month (YYYY-MM). Returns invoice count, total revenue, average invoice value, and unique customers per month.

📄 **Tables Used:** invoice

🔑 **Key Columns:**
year_month, invoice_count,
total_revenue,
avg_invoice_value,
unique_customers

```
-- Group invoices by month and aggregate
SELECT DATE_FORMAT(invoice_date, '%Y-%m') AS year_month,
COUNT(DISTINCT invoice_id) AS invoice_count,
SUM(total) AS total_revenue,
AVG(total) AS avg_invoice_value,
COUNT(DISTINCT customer_id) AS unique_customers
FROM invoice
GROUP BY DATE_FORMAT(invoice_date, '%Y-%m')
```

Why This Matters: Track revenue trends, seasonality, and customer acquisition over time. Essential for monthly reporting and forecasting.

3

vw_sales_by_genre

Genre Performance

Business Question: Which music genres drive the most revenue and sales volume?

Returns genre-level revenue, units sold, invoice count, and revenue percentage of total. Helps identify which genres are most profitable.

 **Tables Used:** invoice, invoice_line, track, genre

 **Key Columns:**
genre_name,
total_units_sold,
total_revenue,
revenue_percentage

```
-- Join to genre and calculate revenue percentage
SELECT g.genre_name,
SUM(il.quantity) AS total_units_sold,
SUM(il.unit_price * il.quantity) AS total_revenue,
ROUND( SUM(il.unit_price * il.quantity) / (SELECT
SUM(unit_price * quantity) FROM invoice_line) * 100, 2 ) AS
revenue_percentage
FROM invoice i JOIN invoice_line il ON i.invoice_id =
il.invoice_id
JOIN track t ON il.track_id = t.track_id
JOIN genre g ON t.genre_id = g.genre_id
GROUP BY g.genre_id, g.genre_name
```

Why This Matters: Guides inventory decisions, marketing focus, and product strategy. Helps understand customer preferences by genre.

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vw_sales_by_country

Geographic Revenue

Business Question: Which countries are our best-performing markets?

Groups invoices by billing country. Returns revenue, invoice count, average invoice value, and customer count per country.

📄 **Tables Used:** invoice

🔑 **Key Columns:**

billing_country, invoice_count,
total_revenue,
avg_invoice_value,
customer_count

```
-- Group by country and aggregate
SELECT billing_country,
COUNT(DISTINCT invoice_id) AS invoice_count,
SUM(total) AS total_revenue,
AVG(total) AS avg_invoice_value,
COUNT(DISTINCT customer_id) AS customer_count
FROM invoice
GROUP BY billing_country
```

Why This Matters: Identifies expansion opportunities, guides marketing spend allocation, and reveals regional customer preferences.

5

vw_customer_lifetime_value

Customer Value & Recency

Business Question: Who are our most valuable customers, and when did they last purchase?

Calculates total spend, invoice count, average invoice, last purchase date, days since last purchase, and spending rank for each customer.

 **Tables Used:** customer, invoice

 **Key Columns:**
customer_id, total_spent,
avg_invoice,
days_since_last_purchase,
spending_rank

```
-- Uses window function to rank customers by spend
SELECT c.customer_id,
COUNT(DISTINCT i.invoice_id) AS total_invoices,
SUM(i.total) AS total_spent,
AVG(i.total) AS avg_invoice,
MAX(i.invoice_date) AS last_purchase_date,
DATEDIFF(CURDATE(), MAX(i.invoice_date)) AS
days_since_last_purchase,
RANK() OVER (ORDER BY SUM(i.total) DESC) AS spending_rank
```

Why This Matters: Identifies VIP customers for loyalty programs, flags churn risk, and enables targeted re-engagement campaigns.

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vw_top_customers

Top 10 VIPs

Business Question: Who are our top 10 customers by lifetime value?

Filters the CLV view to return the top 10 highest-spending customers with full details.

Tables Used:

vw_customer_lifetime_value

Key Columns:

customer_id, name, country,
total_spent, invoices,
last_purchase_date

```
-- Simple filter on the CLV view
SELECT * FROM vw_customer_lifetime_value
WHERE total_spent IS NOT NULL
ORDER BY total_spent DESC
LIMIT 10
```

Why This Matters: VIP treatment, personalized marketing, referral programs, and high-value customer retention.

Business Question: Which tracks generate the most revenue and sell the most units?

Rank tracks by revenue and quantity. Returns track details with artist, album, genre, purchase count, and both revenue and quantity ranks.

 **Tables Used:** track, invoice_line, album, artist, genre

 **Key Columns:** track_name, artist_name, genre_name, total_revenue, revenue_rank, quantity_rank

```
-- Dual ranking by revenue and quantity using window functions
SELECT t.track_id, t.track_name, a.artist_name,
SUM(il.quantity) AS total_quantity_sold,
SUM(il.unit_price * il.quantity) AS total_revenue,
RANK() OVER (ORDER BY SUM(il.unit_price * il.quantity) DESC)
AS revenue_rank,
RANK() OVER (ORDER BY SUM(il.quantity) DESC) AS quantity_rank
```

Why This Matters: Identifies best-selling products, guides inventory decisions, and reveals popular artists/genres that customers prefer.

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vw_employee_performance

Sales Agent KPIs

Business Question: Which sales agents are performing best?

Returns sales agent metrics: customers served, invoices handled, total revenue generated, average invoice value, and revenue rank.

 **Tables Used:** employee, customer, invoice

 **Key Columns:** first_name, last_name, customers_served, total_revenue_generated, revenue_rank

```
-- Only sales agents, ranked by revenue
SELECT e.employee_id, e.first_name, e.last_name,
COUNT(DISTINCT c.customer_id) AS customers_served,
COUNT(DISTINCT i.invoice_id) AS invoices_handled,
SUM(i.total) AS total_revenue_generated,
RANK() OVER (ORDER BY SUM(i.total) DESC) AS revenue_rank
FROM employee e
LEFT JOIN customer c ON e.employee_id = c.support_rep_id
LEFT JOIN invoice i ON c.customer_id = i.customer_id
WHERE e.title LIKE '%Sales%'
```

Why This Matters: Performance evaluation, incentive planning, training needs identification, and best practice sharing among agents.

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vw_sales_trend

Daily Sales with Rolling Averages

Business Question: How is revenue trending day by day?

Returns daily revenue with 7-day rolling total, 7-day rolling average, and cumulative revenue from the first day.

📄 **Tables Used:** invoice

🔑 **Key Columns:**
invoice_date, daily_revenue,
rolling_7_day_avg,
cumulative_revenue

```
-- Window functions for rolling and cumulative calculations
SELECT invoice_date,
total AS daily_revenue,
SUM(total) OVER (ORDER BY invoice_date ROWS BETWEEN 6
PRECEDING AND CURRENT ROW) AS rolling_7_day_total,
AVG(total) OVER (ORDER BY invoice_date ROWS BETWEEN 6
PRECEDING AND CURRENT ROW) AS rolling_7_day_avg,
SUM(total) OVER (ORDER BY invoice_date) AS cumulative_revenue
```

Why This Matters: Short-term performance monitoring, early warning for sales dips, and understanding overall revenue trajectory.

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vw_album_performance

Album Revenue Ranking

Business Question: Which albums generate the most revenue?

Ranks albums by revenue. Returns album details with artist, track count, units sold, total revenue, and revenue rank.

📋 **Tables Used:** album, artist, track, invoice_line

🔑 **Key Columns:** album_title, artist_name, track_count, total_revenue, revenue_rank

```
-- Album-level aggregation with ranking
SELECT al.album_id, al.album_title, a.artist_name,
COUNT(DISTINCT t.track_id) AS track_count,
SUM(il.quantity) AS total_units_sold,
SUM(il.unit_price * il.quantity) AS total_revenue,
RANK() OVER (ORDER BY SUM(il.unit_price * il.quantity) DESC)
AS revenue_rank
```

Why This Matters: Identifies successful album releases, guides marketing for similar artists, and reveals merchandise/bundle opportunities.

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vw_artist_revenue

Artist Revenue Ranking

Business Question: Which artists generate the most revenue?

Returns artist-level revenue, album count, and units sold.
Helps understand which artists drive the most business.

 **Tables Used:** artist,
album, track, invoice_line

 **Key Columns:**
artist_name, album_count,
total_units_sold,
total_revenue

```
-- Artist-level aggregation
SELECT a.artist_id, a.artist_name,
COUNT(DISTINCT al.album_id) AS album_count,
SUM(il.quantity) AS total_units_sold,
SUM(il.unit_price * il.quantity) AS total_revenue
FROM artist a
JOIN album al ON a.artist_id = al.artist_id
JOIN track t ON al.album_id = t.album_id
JOIN invoice_line il ON t.track_id = il.track_id
```

Why This Matters: Artist contract negotiations, marketing budget allocation, and identifying rising artists for investment.

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vw_invoice_summary

Time-Based Aggregation

Business Question: What are our revenue and invoice metrics by year, quarter, and month?

Returns aggregated metrics grouped by year, quarter, and month for high-level reporting.

📄 **Tables Used:** invoice

🔑 **Key Columns:** year,
quarter, month,
invoice_count, total_revenue,
avg_revenue

```
-- Extract time components and aggregate
SELECT YEAR(invoice_date) AS year,
QUARTER(invoice_date) AS quarter,
MONTH(invoice_date) AS month,
COUNT(*) AS invoice_count,
SUM(total) AS total_revenue,
AVG(total) AS avg_revenue
```

Why This Matters: Quarterly business reviews, year-over-year comparisons, and seasonal planning for peak periods.

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vw_repeat_purchase_rate

Customer Loyalty

Business Question: What percentage of customers make repeat purchases?

Calculates overall and country-level repeat purchase rates. Returns total customers, repeat customers, and repeat rate percentage.

 **Tables Used:** invoice, customer

 **Key Columns:** segment (Overall/Country), total_customers, repeat_customers, repeat_rate_percent

```
-- Uses UNION for overall + country-level
-- Overall: count customers with >1 invoice
-- By Country: grouped by country
SELECT 'Overall' AS segment,
COUNT(DISTINCT customer_id) AS total_customers,
SUM(CASE WHEN purchase_count > 1 THEN 1 ELSE 0 END) AS
repeat_customers,
ROUND(SUM(CASE WHEN purchase_count > 1 THEN 1 ELSE 0 END) /
COUNT(DISTINCT customer_id) * 100, 2) AS repeat_rate_percent
```

Why This Matters: Measures customer loyalty, identifies high-loyalty markets, and guides retention strategy and campaigns.

Business Question: How can we group customers for targeted marketing?

Classifies customers into segments based on Frequency (purchase count), Monetary (total spend), and Recency (days since last purchase).

 **Tables Used:** customer, invoice

 **Key Columns:**
customer_id, frequency, monetary, recency_days, customer_segment (e.g., "Loyal / High Spender")

```
-- Uses CASE statements for segmentation logic
-- Frequency: Inactive, Churned, One-Time, Regular, Loyal
-- Monetary: None, Low, Medium, High Spender
CASE
WHEN COUNT(DISTINCT i.invoice_id) = 1 AND DATEDIFF(CURDATE(),
MAX(i.invoice_date)) > 180 THEN 'Churned'
WHEN COUNT(DISTINCT i.invoice_id) BETWEEN 2 AND 5 THEN
'Regular'
WHEN COUNT(DISTINCT i.invoice_id) > 5 THEN 'Loyal'
```

Why This Matters: Enables targeted marketing campaigns, personalized promotions, and efficient resource allocation for different customer tiers.

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vw_churn_risk

Churn Prediction

Business Question: Which customers are at risk of churning?

Classifies customers into High (6+ months), Medium (3-6 months), and Low (0-3 months) churn risk based on days since last purchase.

 **Tables Used:** customer, invoice

 **Key Columns:**
customer_id,
last_purchase_date,
days_since_last_purchase,
churn_risk_level

```
-- Categorize by days since last purchase
SELECT c.customer_id, c.first_name, c.last_name,
MAX(i.invoice_date) AS last_purchase_date,
DATEDIFF(CURDATE(), MAX(i.invoice_date)) AS
days_since_last_purchase,
CASE
WHEN DATEDIFF(CURDATE(), MAX(i.invoice_date)) > 180 THEN
'High Risk (6+ months)'
WHEN DATEDIFF(CURDATE(), MAX(i.invoice_date)) > 90 THEN
'Medium Risk (3-6 months)'
ELSE 'Low Risk (0-3 months)'
END AS churn_risk_level
```

Why This Matters: Enables proactive re-engagement campaigns, customer win-back strategies, and reduces revenue leakage from churned customers.

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vw_sales_by_media_type

Format Preference

Business Question: Which media formats are most popular?

Returns revenue and units sold by media type (MPEG audio, AAC, etc.) with revenue percentage of total.

 **Tables Used:** invoice, invoice_line, track, media_type

 **Key Columns:** media_type_name, total_units_sold, total_revenue, revenue_percentage

```
-- Group by media type
SELECT mt.media_type_name,
SUM(il.quantity) AS total_units_sold,
SUM(il.unit_price * il.quantity) AS total_revenue,
ROUND(SUM(il.unit_price * il.quantity) / (SELECT
SUM(unit_price * quantity) FROM invoice_line) * 100, 2) AS
revenue_percentage
FROM media_type mt
JOIN track t ON mt.media_type_id = t.media_type_id
JOIN invoice_line il ON t.track_id = il.track_id
```

Why This Matters: Prioritizes popular formats, phases out low-performing formats, and optimizes file storage and delivery decisions.

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vw_monthly_new_customers

Acquisition Trends

Business Question: How many new customers are we acquiring each month?

Returns the number of new customers acquired per month based on their first purchase date.

📋 **Tables Used:** invoice

🔑 **Key Columns:**
acquisition_month,
new_customers

```
-- First purchase date per customer, then group by month
WITH first_purchase AS (
  SELECT customer_id, MIN(invoice_date) AS first_purchase_date
  FROM invoice
  GROUP BY customer_id
)
SELECT DATE_FORMAT(first_purchase_date, '%Y-%m') AS
acquisition_month,
COUNT(customer_id) AS new_customers
FROM first_purchase
```

Why This Matters: Measures marketing effectiveness, identifies acquisition seasonality, and guides budget allocation for customer acquisition.

Business Question: Which genres have the most loyal customers?

Calculates average invoices per customer by genre – a proxy for genre loyalty. Higher values indicate more loyal customers to that genre.

 **Tables Used:** customer, invoice, invoice_line, track, genre

 **Key Columns:**
genre_name,
total_customers,
total_invoices,
avg_invoices_per_customer,
units_sold

```
-- Loyalty indicator: avg invoices per customer by genre
SELECT g.genre_name,
COUNT(DISTINCT c.customer_id) AS total_customers,
COUNT(DISTINCT i.invoice_id) AS total_invoices,
ROUND(COUNT(DISTINCT i.invoice_id) / COUNT(DISTINCT
c.customer_id), 2) AS avg_invoices_per_customer
FROM customer c
JOIN invoice i ON c.customer_id = i.customer_id
JOIN invoice_line il ON i.invoice_id = il.invoice_id
JOIN track t ON il.track_id = t.track_id
JOIN genre g ON t.genre_id = g.genre_id
```

Why This Matters: Identifies which genres drive customer loyalty, enabling genre-specific loyalty programs and focused retention efforts.